

Monash University: Assessment Cover Sheet

Student name	Du	Wenxing	
School/Campus	clayton	Student's I.D. number	30171857
Unit name	ECE2111 - Signals and systems - S2 2021		
Lecturer's name	Dr. Faezeh Marzbanrad	Tutor's name	Dr. Nikola Zlatanov
Assignment name	Assignment report		Group Assignment: No Note, each student must attach a coversheet
Lab/Tute Class:	Lab/Tute Time: Tue 11-13	Word Count:	890
Due date: 24-10-2021	Submit Date: 24-10-2021	Extension granted ☐	

If an extension of work is granted, specify date and provide the signature of the lecturer/tutor. Alternatively, attach an email printout or handwritten and signed notice from your lecturer/tutor verifying an extension has been granted.

Extension granted until (date):/...../..... Signature of lecturer/tutor:

Late submissions policy	Days late	Penalty applied
Penalties apply to late submissions and may vary between faculties. Please refer to your faculty's late assessment policy for details.		

Patient/client confidentiality: Where a patient/client case study is undertaken a signed [Consent Form](#) must be obtained.

Intentional plagiarism or collusion amounts to cheating under Part 7 of the Monash University (Council) Regulations

Plagiarism: Plagiarism means to take and use another person's ideas and or manner of expressing them and to pass these off as one's own by failing to give appropriate acknowledgement. This includes material from any source, staff, students or the Internet - published and unpublished works.

Collusion: Collusion means unauthorised collaboration on assessable written, oral or practical work with another person. Where there are reasonable grounds for believing that intentional plagiarism or collusion has occurred, this will be reported to the Associate Dean (Education) or nominee, who may disallow the work concerned by prohibiting assessment or refer the matter to the Faculty Discipline Panel for a hearing.

Student Statement:

- I have read the university's Student Academic Integrity [Policy](#) and [Procedures](#)
- I understand the consequences of engaging in plagiarism and collusion as described in Part 7 of the Monash University (Council) [Regulations](#) (academic misconduct).
- I have taken proper care to safeguard this work and made all reasonable efforts to ensure it could not be copied.
- No part of this assignment has been previously submitted as part of another unit/course.
- I acknowledge and agree that the assessor of this assignment may, for the purposes of assessment, reproduce the assignment and:
 - i. provide it to another member of faculty and any external marker; and/or
 - ii. submit to a text matching/originality checking software; and/or
 - iii. submit it to a text matching/originality checking software which may then retain a copy of the assignment on its database for the purpose of future plagiarism checking.
- I certify that I have not plagiarised the work of others or participated in unauthorised collaboration or otherwise breached the academic integrity requirements in the Student Academic Integrity [Policy](#).

Date: 24/10/2021 Signature: Wenxing Du *

Privacy Statement:

For information about how the University deals with your personal information go to <http://privacy.monash.edu.au/guidelines/collection-personal-information.html#enrol>

2111 Assignment report

■ Introduction:

The term gait refers to the way in which a person walks. A whole gait cycle is composed of stance phase and swing phase. And we know that a computer force-sensitive system can be used to quantify gait and measure swing and standing time and variability.

This project gave me some data on the relationship between the VGRF (total vertical ground reaction force under each foot) and time measured by this system. I need to process and analyze these signals to estimate the sequence of stance times, swing times and stride times for each foot from the noisy VGRF data.

■ Data exploration:

In section 3.1, I need to plot the VGRF(N) vs. time(s) for both feet and overlying them with different colors, for any 5 second segment of data, in figure 1.

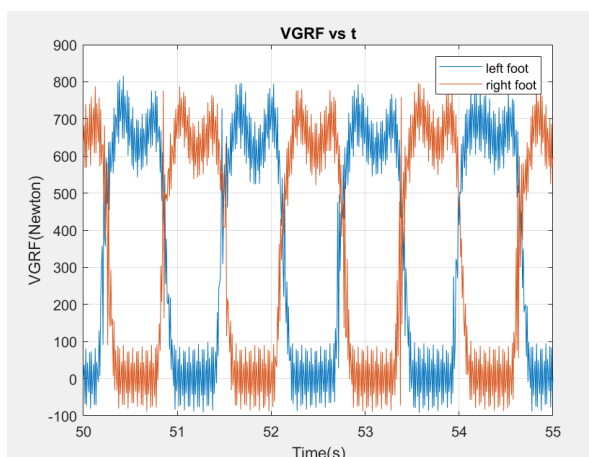


Figure 1

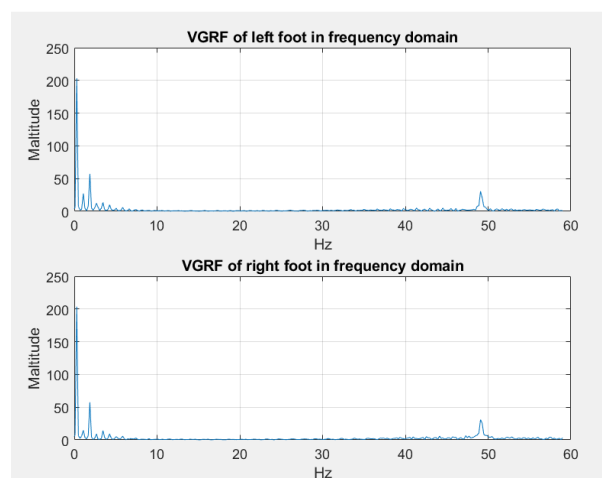


Figure 2

As can be seen from the figure1, the unprocessed signal has a lot of noise, the image shows a lot of burrs and jitter. Signals pass through '0' multiple times, resulting in an incorrect gait count. That is the reason why I can't find a good threshold value from diagram above.

In section1, I need plot the VGRF signal in frequency domain as well, in figure 2. Notice that the negative frequency is meaningless so I delete them.

In section3.2, as we know that the frequency domain FFT plot shows the aggregates of the frequency content over all the gait phases in the signal instead of showing at each gait phase. In order to analyze content of each signal segment, we should divide signal into time intervals short enough to make signal periodic in each section.

As the result, I visualize how the frequency content of the signal changes over time by windowing the signal and estimating the spectrum.

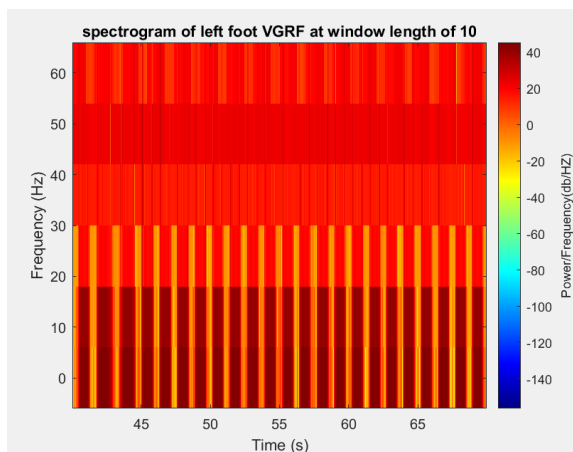


Figure 3

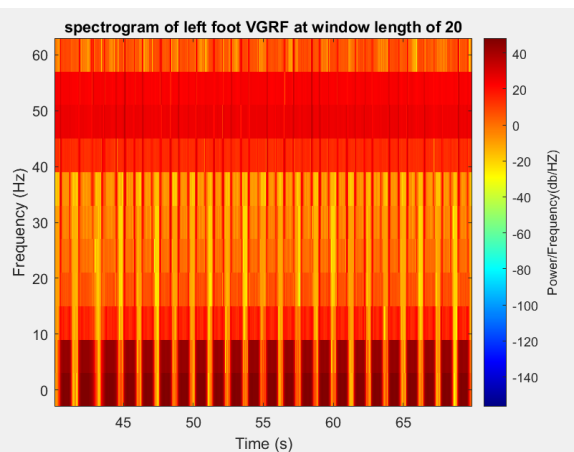


Figure 4

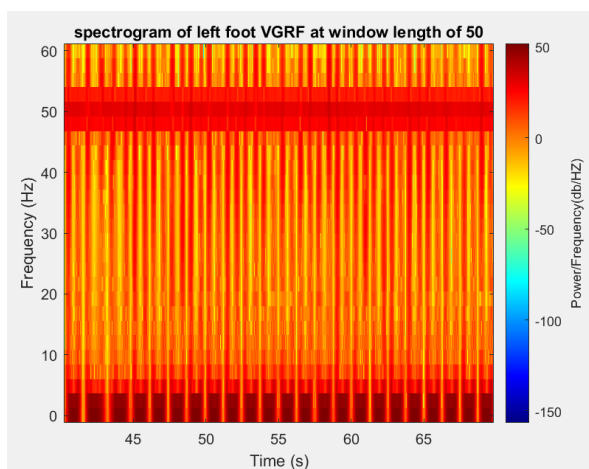
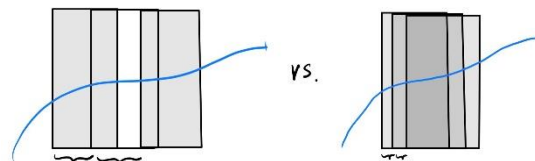


Figure 5



Draft 1

By observing the above three figures, we can see that: when the length of window is between 10 and 50, the larger the length is, the reflection of ordinate more serried, the reflection of frequency more clearly.

The draft 1 is my understanding of the effect of length of window.

The distribution of main noise can already be clearly seen in figure 5. Or I can use three-dimensional images to make it even more obvious:

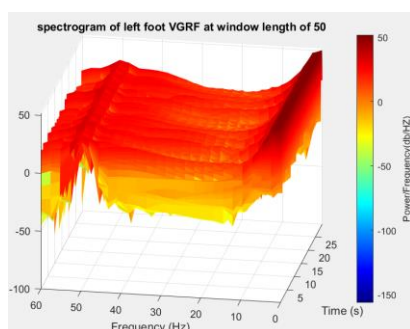


Figure 6

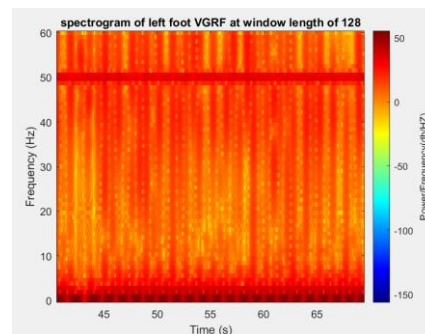


Figure 7

However, it doesn't mean that the larger length of window, the better diagram will display for me. I tried the length of window is 128 and 256, as we can see, in figure 7, when the length of window is too large, the data

graph is divided into small points one by one because of too dense, and no matter on the horizontal or vertical coordinates cannot be clearly seen.

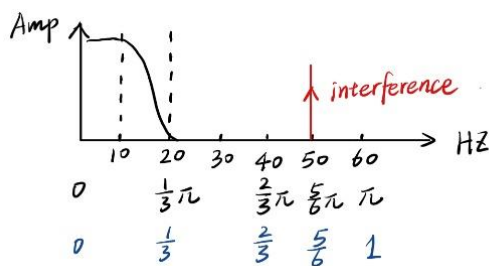
The main noise component from my spectrogram plots is at the position when frequency equals to 50Hz. I think the major source of noise is Australia 's electricity standard frequency, which is 50 Hz and is one of the most common and widespread applications.

In Australia the **standard voltage is 230 V and the frequency is 50 Hz**. You can use your electric appliances in Australia, if the standard voltage in your country is in between 220 - 240 V (as is in the UK, Europe, Australia and most of Asia and Africa).

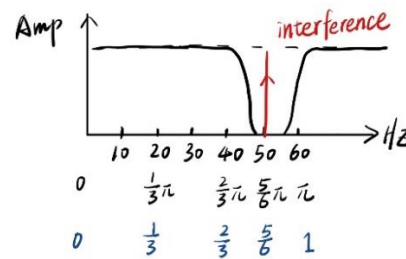
Figure 8

■ Filtering:

I have learnt four types of filters, I choose two of them have a try to compare which one is the best for this project. The low-pass filter attenuates or eliminates frequencies above the specified frequency. The band stop filters are used to remove signal frequencies within a specified range.



Draft 2



Draft 3

The draft 2 and 3 is my process of thinking. I think the low pass filter suitable this question. For matlab code, When order is small , the conv function is a good choice for FIR filter because it can correct delay and minimizes the effect of delay in the output; the firpm function is carried out using the minimum error criterion.

I tried the band stop filter as well, but the filtered diagram still in a mess. It doesn't work very well.

As the result, plotting the filtered VGRF for both feet vs. time (with swing and stance phases marked) , in figure 9:

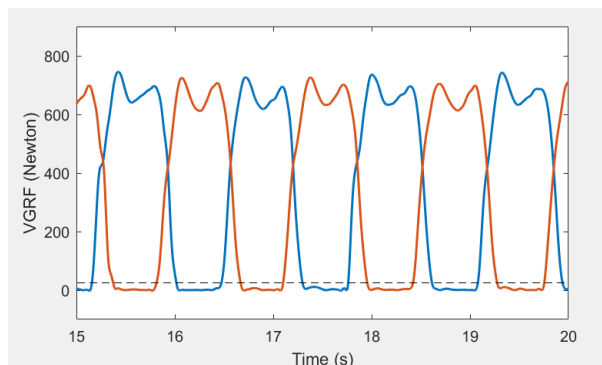


Figure 9

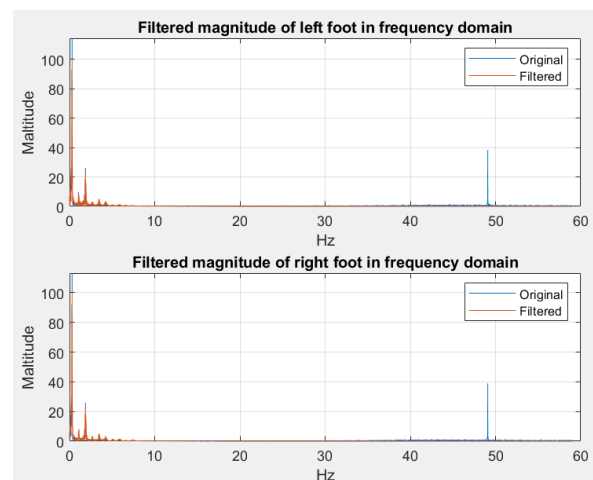


Figure 10

Compare with the figure 1 in section 1, there are no more burrs and noise in figure 9, the lines are very smooth and regular.

Plotting the filtered VGRF signal in frequency domain, in figure 10. The orange line is mixed with the blue line. The blue line is the original signal, and the orange line is the filtered signal. It can be seen that the noise at about 50hz has been removed after filtering.

■ Computing gait parameters :

Mark the onset of swing and stance phases on the filtered VGRF time—domain plot, in figure 11:

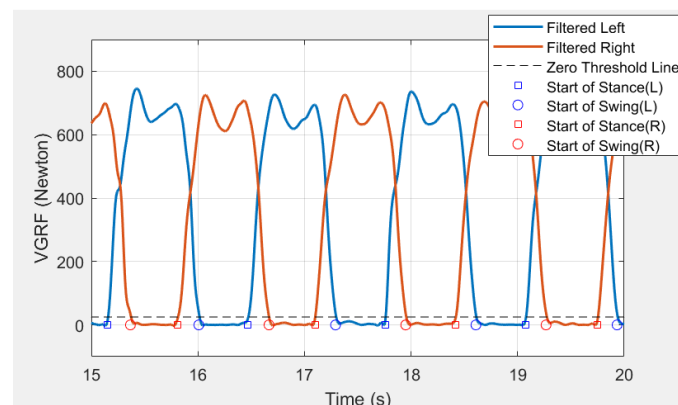


Figure 11

After filtering, MATLAB code : n is the points of signal, when it less than threshold but the $n+1$ larger than threshold, it is the start of stance; when it larger than threshold but the $n+1$ less than threshold, it is the start of swing; the index part which find equals to 1 means heel strike or tiptoe off the ground; the threshold is 25 because if it is larger , the signal will still have plenty of ‘ noise ‘ across the 0.

Plot the stride time vs. cycle, swing time vs. cycle and stance time vs. cycle for each foot, in figure 12:

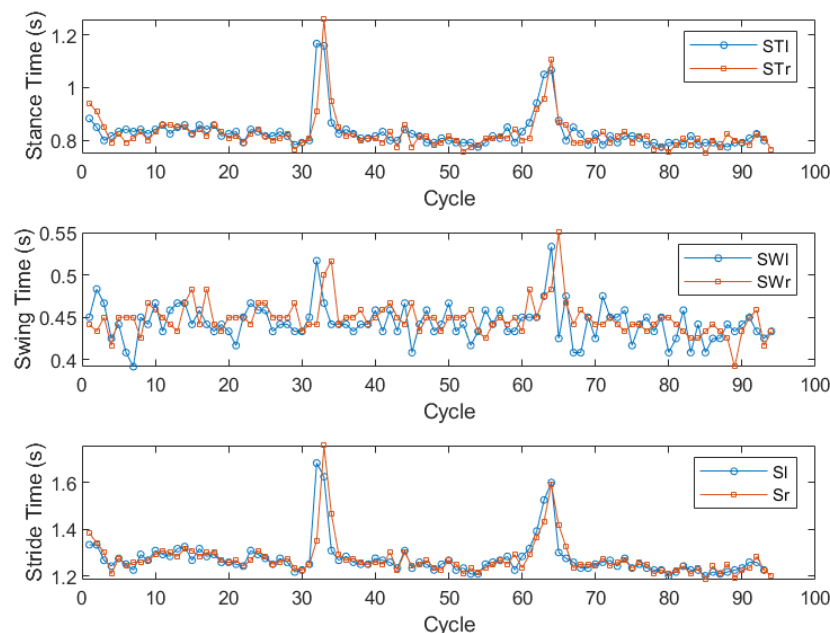


Figure 12

The blue line is for left foot and orange line is for right foot.